NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

ACADEMIC YEAR 2025-2026
SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 1: L1

Question 1 (Lecture 1)

Which of the following statements about agents are true? Select all that apply.

(a) Human and robot agents differ because the latter do not have actuators.

(b) Eyes and ears are examples of a human agent's actuators.

(c) An agent senses its environment through perceptors.

(d) Percepts are the agent's perceptual inputs from the environment at any given time.

(e) Touchless faucets can also be viewed as agents.

Question 2 (Lecture 1)

Which of the following statements regarding rational agents are true? Select all that apply.

(a) Rational agents always know the environment dynamics and can predict future updates of the environment.

(b) A rational agent is omniscient

(c) A rational agent selects an action that maximizes the expected value of its performance measure.

(d) A rational agent is not guaranteed to find optimal solutions.

Question 3 (Lecture 1)

Which of the following are necessary to formally define a problem for a problem-solving agent? Select all that apply.

(a) A clear goal condition that defines when the agent has successfully solved the problem

(b) Specifying the outcome of each action that the agent could take

(c) The initial state from which the agent begins its problem-solving process

(d) The set of possible actions the agent can perform in each state

Python in EE2213 – Getting Started

From Tutorial 2 onwards, we'll be using Python for some computational tasks. To ensure a smooth experience, we recommend installing **Anaconda**, a Python distribution for data processing, analytics, and scientific computing. It includes core packages like NumPy, SciPy, matplotlib, pandas, and scikit-learn, along with development tools such as IPython and Jupyter Notebook. Anaconda is available for Windows, macOS, and Linux, and is ideal if you don't already have these packages installed.

Helpful Resources:

- W3Schools Python Tutorials
- Python Data Types Overview
- NumPy Quickstart Tutorial
- NumPy Tutorial with Jupyter & Colab

Optional Python Clinic Sessions for EE2213

To help you get started, we're offering **Python Clinic Session 1**, covering:

- Setting up Anaconda and Jupyter Notebook
- Data types: list, tuple, dictionary
- Control structures: if-else, for loops, while loops
- Using NumPy and Matplotlib
- Writing and calling functions
- Debugging and understanding error tracebacks
- Assignment guidance

To accommodate different students' schedules, Python Clinic Session 1 will be held online in two separate slots. You can attend either one of the following slots:

- **Slot 1:** Thursday, 21 Aug, 9:00–10:00 PM
- **Slot 2:** Monday, 25 Aug, 3:00–4:00 PM